Number Theory — problem 6.137

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(a) If the positive integers x, y satisfy $2x^2 + x = 3y^2 + y$ then $x - y$, $2x + 2y + 1$, $3x + 3y + 1$ are perfect squares. (PMO 1964/65.)

(b) Find all integral solutions of $2x^2 + x = 3y^2 + y$.